Claim Complexity Algorithm

AI-assisted scoring for worker compensation claim



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September 2026

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I. Introduction

Claims management currently depends on human examiners going over every claim, no two claims demand the same effort or have any quick indicators to determine its complexity right away. Some are routine approvals that close quickly and can be considered simple claims. Others stay open longer due to needed extended reviews, denials, or disputes over treatment which increases complexity. Penser North America currently has no way to tell these apart in advance. Complexity is assessed through examiner judgment, which can vary between individuals. The motivation for this project is to give examiners a quick, consistent gauge of whether a claim will be complex, meaning how long it will take and how much work it will require, along with the cost likely associated with it. We are building a model trained on Penser's last five years of claims to provide that gauge for every claim.

II. Background and Related Work

Workers compensation pays for medical treatment and lost wages when an employee is hurt on the job. Third party companies like Penser manage these claims for employers and insurers, and an examiner handles each claim from the time it is filed until it closes. Each claim has fields the examiner enters.

III. Project Overview

The Problem

There is currently no standardized data driven way to measure how complex a workers compensation claim is. Complexity is estimated subjectively and it depends on many factors like the injury type, industry time loss, return to work status, and vocational rehabilitation needs legal activity fraud or malingering risks either by an injured worker or a medical provider and the settlement trajectory. No single person can easily weigh all these consistencies across 10,000 claims.

Issues

Assignment and staffing: Without a reliable measure, TPAs cannot deliberately match claims to experienced examiners or anticipate staffing needs as their client portfolio grows.

Cost Visibility: Employers clients have limited insight into which claims will become costly which makes it harder to set reserves and design safety programs.

Misleading proxies: Examiner activity, such as the number of touch points, is easy to measure, but it can reflect an individual examiner's habits rather than how difficult the claim really is. Treating it as complexity would build that bias into any model.

Defensibility: Time loss decisions need to be accurate and defensible, and an informal approach makes it harder to justify the claims.

The Goal: The project will deliver an AI-assisted framework that combines claim characteristics into a composite complexity score and a companion cost score for each claim. It is being built first for Washington State claims and designed so it can later expand to other states.

Objectives:

Define and score complexity: Build a score that reflects how long a claim is likely to take and how much work it requires without treating examiner activity alone as complexity.

Estimate cost: Use closed claims with known final costs as anchors then work backward to find which characteristics are associated with the costs.

Allow tuning: Support adjustable weights including an inflation related weight so cost estimates account for changing claim costs.

Communicate Confidence: Shows examiners a 0 – 100 confidence percentage and gives leadership and engineering a graphical view.

Stay interpretable: Make the factors behind each score explainable.

Data:

We will begin with about 10,000 claims from the last five years, roughly 7 MB, covering industries such as agriculture, construction, police, fire, and maritime work. The data includes examiner-entered fields such as claim start date and actual cost, plus narrative diary notes that are scrubbed of identifying information and tokenized before we receive them. Penser will send the column headers first, and the full five-year notes dump is expected around early October because Penser's claims system is moving to a new platform. If the data is not diverse enough or accuracy is too low, Penser can provide older claims going back to the 1990s. Every participant signs a non-disclosure agreement because the data includes legal and medical information.

Desired outcomes and Requirements:

The final system should output a complexity score, a cost score, and a confidence value for each claim. Claims are expected to arrive gradually, about one per minute on average. The system should handle three per minute to cover examiners working at the same time, with an inference ceiling of 10 minutes. The model will be retrained about every six months. Per the proposal, the deliverables are AI/ML integration, a dashboard with visualizations, a backend system, and a database solution. Development starts locally, with AWS or SageMaker evaluated later for hosting. The working budget for AWS, Claude, and related needs is about $300 to $500, and our mentor, Joell Parks, is to be notified if costs get close to $500. A retraining budget will be proposed once retraining is ready.

Intended Users and Expected Impact:

TPAs: assign claims to appropriately experienced staff and plan capacity as their client portfolio grows.

Employers: see which claim types and job classifications drive high complexity outcomes informing safety program and reserve strategy.

Insurance brokers: get better insight into underwriting needs by client industry.

Injured workers: get closer oversight on the claims that need it most.

State of Washington: Also identified as an intended user in the proposal.

The broader outcome is a more accurate and defensible way to calculate claim complexity and time loss turning that awareness into measurable operational and safety results.

Timeline

Work is planned across three sprints: planning and research, initial model training, and retraining or evaluation, which includes checking whether the 10,000 claims are enough. Sprint 1 covers reviewing the column headers and deciding which model approaches fit the data. Sprints are due October 5, November 5, and December 5. The team meets with Penser online every week, and Joell has offered regular stand-ups during the sprints.

IV. Client and Stakeholder Identification and Preferences

Primary Client and Mentors

Our primary client is Penser North America, represented by our mentor and primary contact, Joell Parks. Keno Peck and Ritvik Sharma are additional contacts at Penser. Keno works directly with our team on the data, including the column headers, the claims dataset, and the sanitized notes, and on the development plan. Ritvik is coordinating our Claude access. Joell guides the project, set our working budget of about $300 to $500, and wants to be told if costs get close to $500.

Development Team:

Kevin Herrera, Brian Leung, and Ishaan Joshi are responsible for analyzing claim complexity, building the complexity, cost, and confidence outputs, and checking whether the initial dataset is enough.

Stakeholders and Expected Users

The primary users will be TPAs and claims examiners. They need a quick, consistent read on how complex a claim will be and what it will likely cost, so claims can go to examiners with the right experience and workload can be planned as the client portfolio grows. Examiners will see a 0 to 100 confidence percentage next to each score, which matches Penser's other platforms.

Penser leadership and engineering: They will get a graphical view of the results and will maintain the system, including retraining the model about every six months. Before a retraining budget is set, they want us to propose an approach and cost estimate.

Employer-clients: They want to see which claim types and job classifications lead to high-complexity claims, which helps with safety programs and reserve strategy.

Insurance brokers: They want better insight into underwriting needs based on their clients' industries.

Injured workers: They are affected indirectly. The system aims to bring closer oversight to the claims that need it most. Because the data comes from their claims, identifying details are scrubbed and the notes are tokenized before we receive them, and everyone on the project signs a non-disclosure agreement.

V. Glossary

Data Schema: defines the structure and organization of data within a database. It serves as a blueprint for how data is stored and managed and accessed through the dashboard.

Data Sanitization: Is a process of removing, masking, tokenizing or otherwise transforming data so that sensitive content is not exposed to downstream systems, users, logs, or model outputs.

Complexity Score: should reflect how long a claim may take and the workload required to process it.

Cost Score: The models estimate of a claims cost produced alongside the complexity score as one of two primary outputs.

Complexity Factors: To examine broad factors including claim price and high-level examiner touch points medical conditions total claim cost time loss and return to work vocational rehabilitation and litigation.

Avoiding Examiner Bias: That repeated examiner interactions should not automatically define a claim as complex because excessive touch points may reflect examiner behavior rather than claim characteristics.

Inference Time: The time the model takes to process a claim and produce its score, 10 minutes is the planning ceiling though the model should ideally finish sooner given the expected dataset.

Data Status: When claims data and sanitized notes are ready to be handled.

Third-part Administrator: A company that manages claims on behalf of an employer or insurer including assigning examiners and overseeing claim handling.

Tokenization: Replacing words or data with tokens so the text can be analyzed without exposing the original content.

VI. References